GOVERNMENT OF WESTERN AUSTRALIA

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Department of Mines, Industry Regulation and Safety

248284.2.pdf *Geochronology Record 1908*

248284.2: metamonzogranite, Merino prospect

(Yilgarn Craton granites; Youanmi Terrane, Yilgarn Craton)

IOH Fielding, Y Lu, MTD Wingate and TJ Ivanic

Location and sampling

CORRIGIN SI50-03, NAREMBEEN 2533 Zone 50, MGA 636856E 6440455N
Warox Site TJISWT000026 Sampled on 22 October 2020

This sample was collected from the 72.4 – 74.0 m depth interval of diamond drillcore THDD025, drilled in 2018 by Explaurum Operations Pty Ltd at the Merino prospect of their Tampia gold project (Franey, 2019). The drillhole is located about 5.6 km west of Rodger Road, 2.8 km southwest of Mount Arrowsmith, and 2.2 km north of Tampia Hill.

Geological context

The unit sampled is a metamonzogranite (Fig. 1) within the western Youanmi Terrane of the Yilgarn Craton (Quentin de Gromard et al., 2021). Monazites from this sample yielded an age of metamorphism of 2617 ± 8 Ma (Fielding et al, 2022). A metagranodiorite sampled about 30 km to the north yielded an igneous crystallization age of 2657 ± 5 Ma (GSWA 250432, Wingate et al., 2021). A high-Ca monzogranite sampled about 31 km to the east-northeast yielded U–Pb zircon dates of c. 2680–2630 Ma (Qui and McNaughton, 1999).

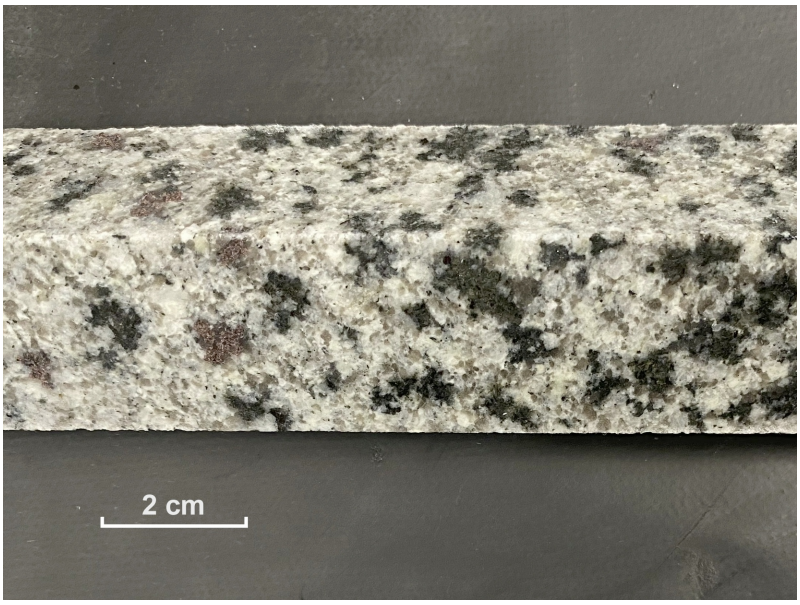

Figure 1. Drillcore image (wet surface) of sample 248284: metamonzogranite, Merino prospect

Petrographic description

The sample is an equigranular, medium-grained, cordierite–orthopyroxene–garnet-bearing metamonzogranite (Fig. 1), consisting of about 30–35% K-feldspar, 30% quartz, 25% plagioclase, 3% myrmekite, 3% biotite, 2% garnet, 1% orthopyroxene, 1% magnetite, <1% cordierite and accessory muscovite, sulfide minerals and zircon. K-feldspar is anhedral to subhedral, up to 4.5 mm long. Larger grains are microperthitic, some exhibit microcline twinning, and many contain round inclusions of quartz and plagioclase. Quartz is anhedral, irregular, up to 6 mm in size, weakly strained, and exhibits smooth, curved boundaries with adjacent feldspar grains resulting in a semipolygonal texture. Plagioclase (andesine, An_{32}) is anhedral to subhedral, equant to slightly tabular, and up to 4 mm in size. Curved lobes of myrmekite occur at the margins of some K-feldspar grains. Biotite forms anhedral skeletal laths up to 1.5 mm long, intergrown with quartz, and smaller bladed laths, some of which contain randomly oriented acicular rutile. Garnet is anhedral, embayed, fractured and up to 4.5

mm in size. Orthopyroxene (hypersthene) is subhedral, equant, up to 1 mm in size and associated with biotite. Cordierite is anhedral and >2 mm in size. Magnetite is anhedral to subhedral, equant, embayed, and up to 0.8 mm in diameter. Muscovite forms anhedral and subhedral laths up to 0.5 mm long and is partly replaced by carbonate. The sample lacks a tectonic foliation, although its semipolygonal texture, abundance of microcline and myrmekite, indicate it has been deformed and partly recovered at moderate metamorphic conditions (at least amphibolite facies, based on the presence of cordierite). The sample retains its igneous texture, and the garnet and orthopyroxene may be primary rather than metamorphic.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and mainly subhedral, and slightly rounded. The crystals are up to 450 µm long, and equant to slightly elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, most zircons exhibit concentric zoning, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

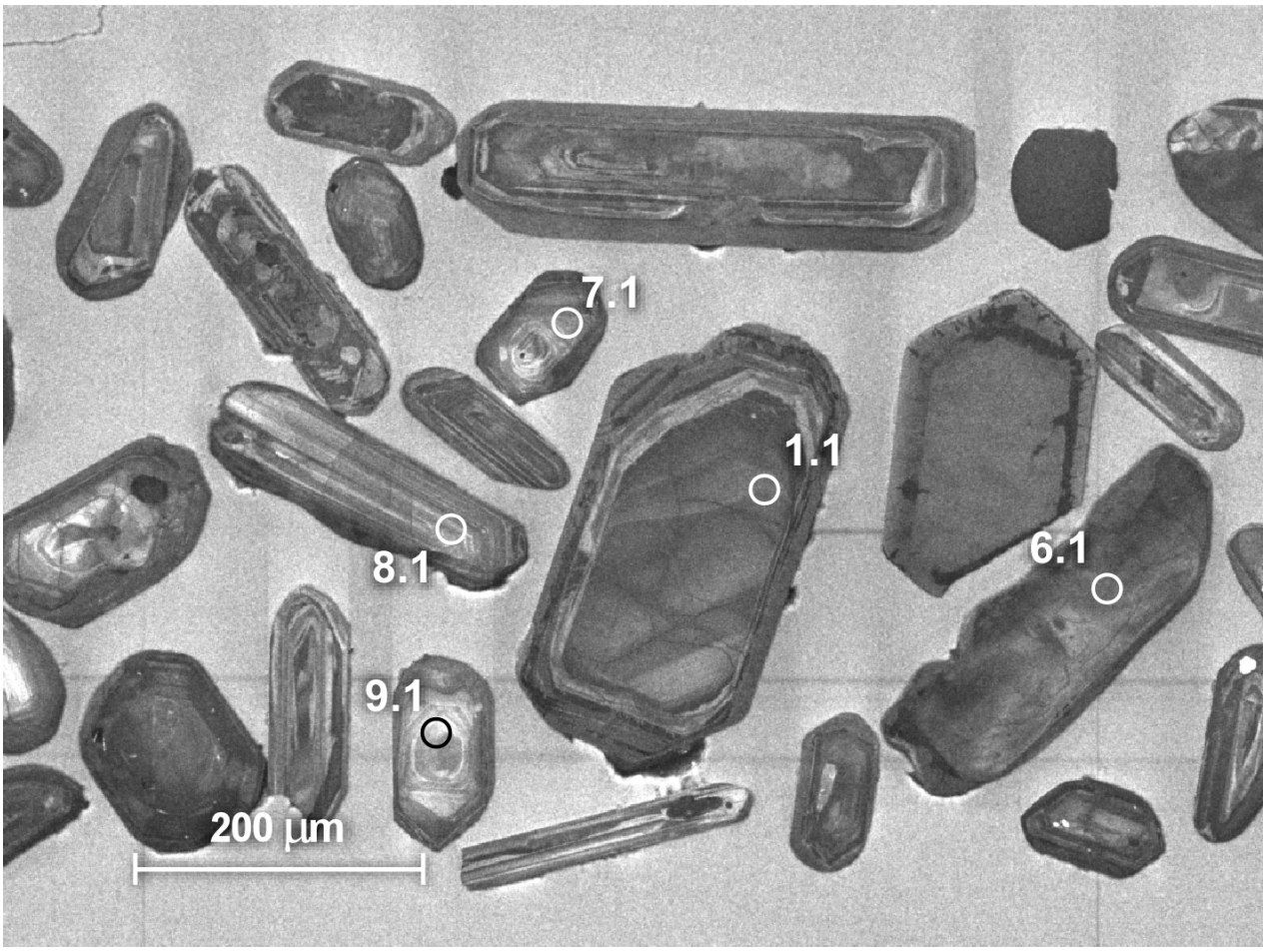

Figure 2. Cathodoluminescence image of representative zircons from sample 248284: metamonzogranite, Merino prospect. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 3–4 June 2021, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.15% (1σ). Isotopic mass fractionation of $^{207}Pb^*/^{206}Pb^*$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.24%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Nineteen analyses were obtained from 18 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Four analyses are >5% discordant. The dates obtained from these four analyses (Group D; Table 1) are unreliable and considered not to be geologically significant. The remaining 15 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses of 15 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2639 ± 7 Ma (MSWD = 2.2).

The date of 2639 ± 7 Ma for the 15 analyses in Group I is interpreted as the igneous crystallization age of the metamonzogranite.

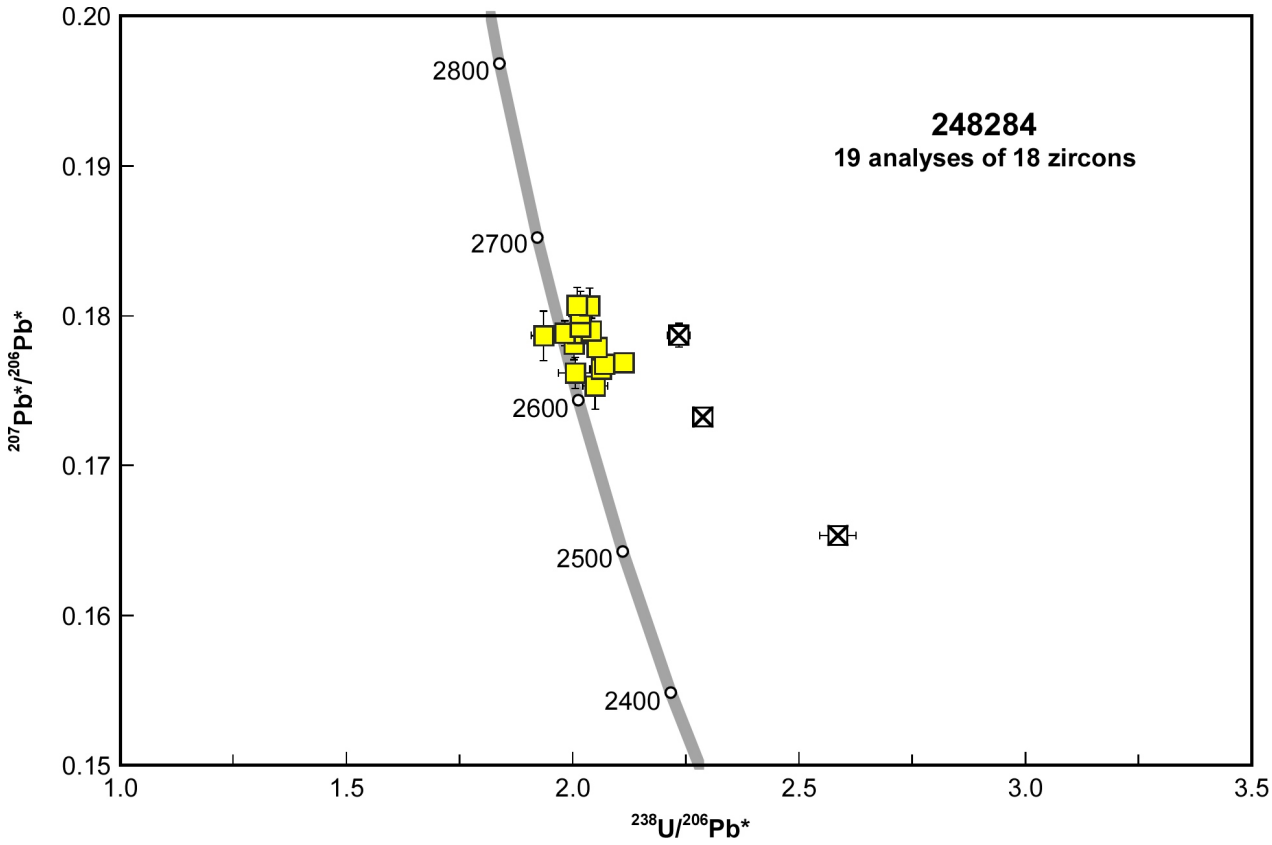

Figure 3. U–Pb analytical data for zircons from sample 248284: metamonzogranite, Merino prospect. Yellow squares indicate Group I (magmatic zircons); crossed squares indicate Group D (discordance >5%). One analysis in Group D is not shown

Links

Table 1. Ion microprobe analytical results for zircons from sample 248284: metamonzogranite, Merino prospect: 248284.2.Table1.xls
SHRIMP spot images: 248284_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

References

Fielding, IOH, Wingate, MTD, Ivanic, TJ and Fisher, CM 2022, 248284: metamonzogranite, Merino prospect; Geochronology Record 1879: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>.

Franey, D 2019, Tampia Gold Project EIS co-funded Deep Diamond Drilling (R#16); Explaurum Operations Pty Ltd: Geological Survey of Western Australia, Statutory mineral exploration report A119888, 18p.

Qiu, YM and McNaughton, NJ 1999, Source of Pb in orogenic lode-gold mineralisation: Pb isotope constraints from deep crustal rocks from the southwestern Archaean Yilgarn Craton, Australia: Mineralium Deposita, v. 34, p. 366–381.

Quentin de Gromard, R, Ivanic, TJ and Zibra, I 2021, Pre-Mesozoic interpreted bedrock geology of the southwest Yilgarn, 2021, in Accelerated Geoscience Program extended abstracts: Geological Survey of Western Australia, Record 2021/4, p. 122–144.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

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